

# Unit 1: Introduction & Understanding Data – 1

## Handwritten Student Notebook

### Syllabus Topics Covered:

OFFICIAL SYLLABUS TOPICS COVERED Introduction & Fundamentals: Need for Machine Learning, Classical Programming vs ML Paradigm, Machine Learning in Relation to Other Fields (AI, Data Mining, Statistics), Challenges of Machine Learning. Machine Learning Explained: Tom Mitchel...

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# Need for Machine Learning

## § 1

In traditional computing (classical programming), a human programmer formulates explicit, deterministic rules and encodes them into software. While this paradigm excels at well-structured logic (such as calculating payroll or sorting a database), it completely breaks down when confronted with high-dimensional, ambiguous real-world phenomena such as recognizing handwritten digits, translating speech, or detecting fraud.

### Definition: Classical Programming vs. Machine Learning:

**Classical Programming:** Humans provide *Rules (Algorithms)* and *Data* to a computer, which runs them to generate *Answers (Outputs)*.

**Machine Learning:** Humans provide *Data* and observed *Answers (Labels/Outputs)* to a computer, which uses mathematical optimization to synthesize the underlying *Rules (Models)*.

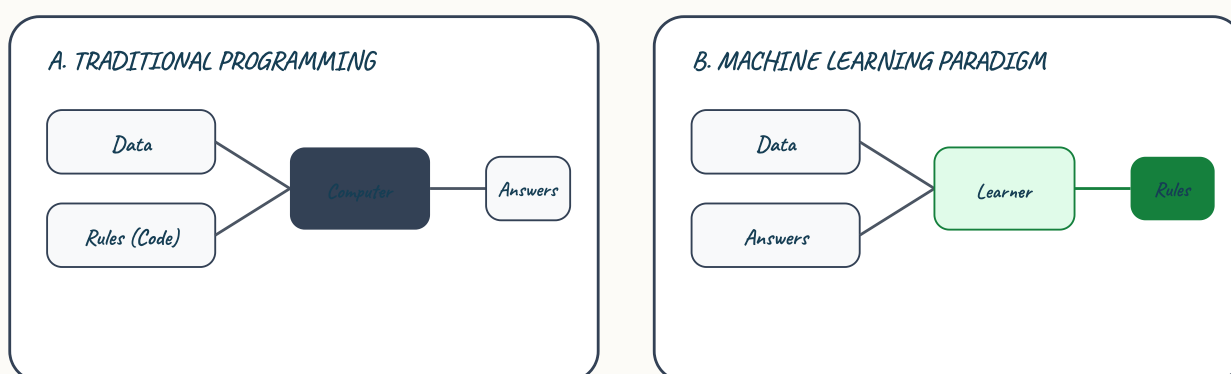


Figure 1.1: Operational inversion from deterministic programming to empirical model induction.

### Drivers of the Need for ML

- **Data Proliferation:** Billions of connected IoT sensors, transactional logs, and mobile devices generate massive data streams too fast for manual curation.
- **Intractable Problem Complexity:** Human expertise is incapable of writing *if – else* conditions to delineate visual concepts like "faces" across diverse lighting, angles, and occlusions.
- **Dynamism and Concept Drift:** Environments evolve over time (e.g., spam tactics evolve weekly). Hardcoded systems degrade, whereas ML algorithms automatically adapt to new training distributions.

- **Scale and Personalization:** E-commerce engines cannot manually hand-curate product recommendations for 100 million distinct individual consumer preferences.

### Quick Recall

Machine learning shifts programming from *prescribing procedural logic* to *optimizing inductive parameters* directly from empirical observations.

## Machine Learning Explained: Formal Definition

### § 2

While colloquially referred to as "computers learning from experience," academic machine learning requires a rigorous, testable definition. The gold-standard formal definition in computer science is formulated by **Prof. Tom M. Mitchell** (Carnegie Mellon University, 1997).

#### Mitchell's Formal Definition of Learning:

A computer program is said to **learn** from experience **E** with respect to some class of tasks **T** and performance measure **P**, if its performance at tasks in **T**, as measured by **P**, improves with experience **E**.

### Deconstructing the $E, T, P$ Triad with Core Examples

In university examinations, you will frequently be asked to formally specify the  $T, P, E$  triplet for concrete real-world engineering problems:

| Problem Domain                      | Task ( $T$ )                                              | Performance Measure ( $P$ )                                              | Training Experience ( $E$ )                                                          |
|-------------------------------------|-----------------------------------------------------------|--------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Checkers Learning Program           | Playing checkers games against an opponent                | Percent of games won in a championship tournament                        | Playing practice games against itself or recorded historical matches                 |
| Handwritten Digit Recognition       | Recognizing and classifying 10 handwritten digits (0–9)   | Percentage of digits correctly classified on unseen test data (Accuracy) | Database of human-verified scanned handwritten digits with true labels (e.g., MNIST) |
| Autonomous Driving (ALVINN / Waymo) | Driving on four-lane public highways using vision sensors | Average distance traveled before human intervention / disengagement      | Sequence of camera images recorded while observing a human driver steer              |

| Problem Domain    | Task ( $T$ )                                           | Performance Measure ( $P$ )                                   | Training Experience ( $E$ )                                    |
|-------------------|--------------------------------------------------------|---------------------------------------------------------------|----------------------------------------------------------------|
| Email Spam Filter | Classifying incoming email as Spam or Legitimate (Ham) | Precision, Recall, and $F_1$ -score on incoming email traffic | Collection of historical emails marked as spam or ham by users |

### Common Student Mistake:

Never confuse Task ( $T$ ) with Performance ( $P$ ).  $T$  is the *action executed by the system* (e.g., classifying an image).  $P$  is the *quantitative metric evaluated over test trials* (e.g., accuracy, mean squared error).

### Quick Recall

Any ML problem must be formalizable into Mitchell's triad: **Task ( $T$ )** to perform, **Performance metric ( $P$ )** to optimize, and **Experience ( $E$ )** observed.

## Machine Learning in Relation to Other Fields

### § 3

Machine learning sits at the confluence of several foundational computational and quantitative disciplines. It borrows theoretical constructs from mathematics and applies them within software systems.

| Discipline                   | Primary Focus                                                                                                   | Interconnection with Machine Learning                                                                                                                                                                                                      |
|------------------------------|-----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Artificial Intelligence (AI) | Creating broad computational agents capable of rational thought and human-like problem solving.                 | ML is a <i>proper subset</i> of AI. While early AI relied on symbol manipulation and expert systems, modern AI is almost entirely driven by statistical learning models.                                                                   |
| Statistics                   | Drawing formal inferences, hypothesis testing, and confidence intervals from samples to understand populations. | Statistics provides the mathematical backbone (probability theory, variance estimation, distributions). Statistics focuses on <i>inference and explainability</i> ; ML emphasizes <i>generalizable predictive accuracy</i> on unseen data. |

| Discipline          | Primary Focus                                                                                          | Interconnection with Machine Learning                                                                                                                    |
|---------------------|--------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data Mining (KDD)   | Discovering novel, hidden, and actionable patterns within massive, heterogeneous datasets.             | Data mining uses ML algorithms as tools to extract rules, but also encompasses data warehousing, OLAP cubes, ETL pipelines, and business visualization.  |
| Pattern Recognition | Classifying raw physical sensory signals (audio waveforms, pixel matrices) into structured categories. | Historically rooted in engineering; modern pattern recognition algorithms (e.g., speech recognition) are implemented using machine learning classifiers. |

### Quick Recall

Statistics proves theoretical bounds and relationships; Computer Science ensures computational efficiency; Data Mining explores large databases; and AI integrates the learned models into autonomous systems.

## Types of Machine Learning Paradigms

### § 4

Machine learning models are broadly categorized according to the nature of the training signal (supervision) provided during the learning phase.

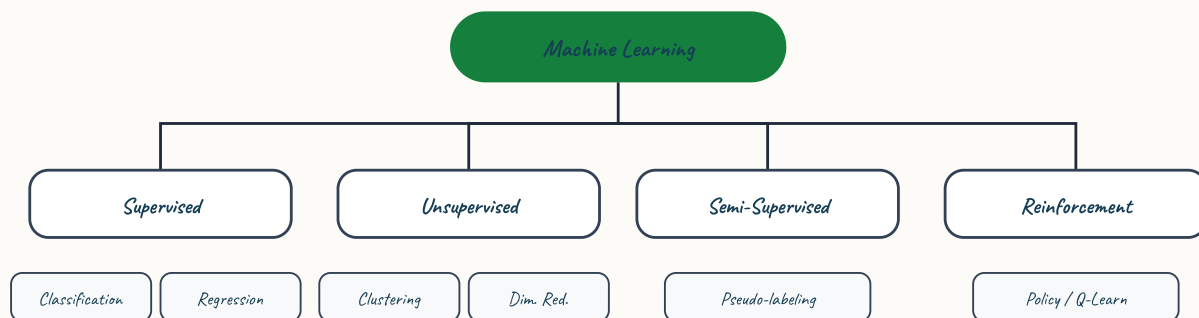


Figure 1.2: Taxonomy and major algorithmic subdivisions of machine learning paradigms.

### 1. Supervised Learning

The learner receives training data formatted as feature-label tuples:  $D = \{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$ , where  $x_i \in \mathbb{R}^d$  is the feature vector and  $y_i$  is the known target ground-truth label. The goal is to induce a mapping function  $f : X \rightarrow Y$  that accurately predicts unseen  $y$  for future  $x$ .

- **Classification:** The label  $y$  is discrete and categorical (e.g., Diagnosing Tumors: Malignant vs. Benign, Spam detection).
- **Regression:** The label  $y$  is continuous and numerical (e.g., Predicting real-estate property valuations, Stock price forecasting).

## 2. Unsupervised Learning

The learner receives only unlabelled instances:  $D = \{x_1, x_2, \dots, x_n\}$  with *no corresponding ground truth  $y$* . The system must autonomously uncover latent intrinsic structures, density distributions, or groupings.

- **Clustering:** Grouping instances such that intra-cluster distance is minimized while inter-cluster distance is maximized (e.g., K-Means, Customer market segmentation).
- **Dimensionality Reduction:** Compressing high-dimensional feature spaces while preserving essential variance (e.g., PCA, t-SNE).
- **Association Rule Mining:** Identifying co-occurrence relationships (e.g., Market Basket Analysis: "Customers who buy bread also buy butter").

## 3. Semi-Supervised Learning

Addresses real-world industrial settings where unlabelled data is cheap and abundant, but human label acquisition is prohibitively expensive (e.g., medical CT scans). The dataset contains a small fraction of labeled examples  $D_L = \{(x_i, y_i)\}_{i=1}^l$  and a massive volume of unlabelled examples  $D_U = \{x_j\}_{j=l+1}^{l+u}$  with  $l \ll u$ .

## 4. Reinforcement Learning (RL)

Based on behavioral psychology. An autonomous **Agent** interacts with an external, dynamic **Environment** through trial and error. At each discrete timestep  $t$ , the agent observes an environmental **State** ( $s_t$ ), selects an **Action** ( $a_t$ ), transitions to state  $s_{t+1}$ , and receives a scalar numerical **Reward** ( $r_t$ ). The goal is to learn an optimal mapping policy  $\pi^*(s)$  that maximizes cumulative discounted future rewards:

$$R_t = \sum_{k=0}^{\infty} \gamma^k r_{t+k+1}, \quad \text{where } \gamma \in [0, 1) \text{ is the discount factor.}$$

Quick Recall

**Supervised:** Teacher provides  $(x, y)$ . **Unsupervised:** Only  $x$  provided, discover patterns.

**Reinforcement:** Agent earns rewards/penalties via environment interaction.

## *Challenges of Machine Learning*

Deploying ML in production settings introduces failure modes that do not exist in conventional software development. These stem from issues with data quantity, data quality, and model parameterization:

| Challenge                             | Symptom & Cause                                                                                                                                                                              | Mitigation Strategy                                                                                                          |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>Overfitting</b><br>(High Variance) | Model memorizes training sample noise and random fluctuations. Achieves near 0% training error but exhibits massive error on unseen test data.                                               | Regularization ( $L_1$ Lasso, $L_2$ Ridge), Dropout, Pruning decision trees, Cross-Validation, Gathering more training data. |
| <b>Underfitting</b><br>(High Bias)    | Model is too simplistic to capture the underlying non-linear relationship (e.g., fitting a straight line to sinusoidal data). Both training and test errors are high.                        | Increase model capacity/complexity, Engineer polynomial features, Reduce regularization constraints.                         |
| <b>Curse of Dimensionality</b>        | As feature dimensions $d$ grow, the volume of space expands exponentially, causing data points to become extremely sparse. Distances between all pairs of points converge to the same value. | Feature selection (Filter/Wrapper techniques) and Dimensionality reduction (PCA, LDA, SVD).                                  |
| <b>Poor Data Quality &amp; Noise</b>  | Data contains sensor errors, typos, missing entries, and outlier anomalies that corrupt the objective loss surface.                                                                          | Robust data cleaning, Median/Mean imputation, Boxplot outlier fencing, Winsorization.                                        |
| <b>Data Drift / Concept Drift</b>     | The statistical properties of the target concept or independent features change over time ( $P_{train}(X, y) \neq P_{deploy}(X, y)$ ).                                                       | Continual learning pipelines, Sliding window retraining, Real-time distribution shift monitoring.                            |

#### Quick Recall

**Overfitting:** Low train error, high test error (memorization). **Underfitting:** High train error, high test error (oversimplification). **Curse of dimensionality:** High dimensions cause sparse distances.

## The Machine Learning Process Lifecycle

Building a machine learning solution is an iterative, structured engineering pipeline rather than a single coding step. In the literature, this is standardized under frameworks such as CRISP-DM (Cross-Industry Standard Process for Data Mining).

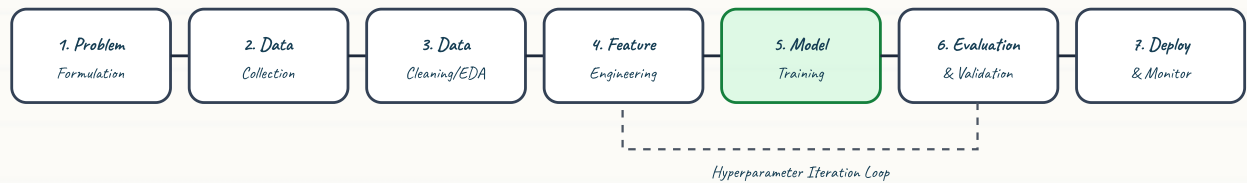


Figure 1.3: End-to-end Machine Learning Lifecycle with iterative refinement loops.

### Detailed Steps of the Lifecycle

1. **Problem Understanding & Formulation:** Mapping a business requirement into an ML task (e.g., determining whether the target requires binary classification, ranking, or regression).
2. **Data Collection & Ingestion:** Querying relational databases, pulling telemetry logs, crawling web content, or acquiring sensor streams.
3. **Data Preprocessing & Cleaning:** Handling corrupt records, treating missing null values via statistical imputation, identifying outliers, and encoding categoricals.
4. **Feature Engineering & Scaling:** Constructing high-signal domain features (ratios, date-time differentials) and standardizing numeric ranges via Min-Max or Z-Score.
5. **Model Selection & Training:** Partitioning data into Train/Validation/Test splits. Optimizing objective loss functions across candidate algorithms (e.g., Logistic Regression, Decision Trees, SVM).
6. **Evaluation & Diagnostics:** Assessing generalization performance on held-out test data using rigorous metrics ( $F_1$ -Score, ROC-AUC, RMSE) and diagnosing bias-variance balance.
7. **Deployment, Serving & Monitoring:** Wrapping models in REST APIs (e.g., FastAPI, Docker), managing inference latency, and tracking real-time prediction drift.

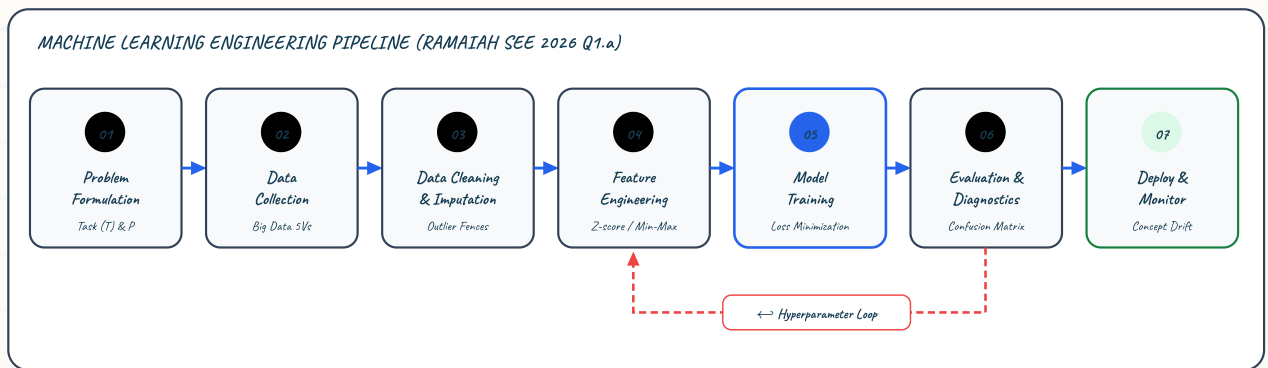


Figure 1.2: End-to-End Machine Learning Process Engineering Lifecycle (Ramaiah SEE 2026 Q1.a Solved)

7-stage production architecture: Problem Formulation → Data Collection → Cleaning & Imputation → Feature Engineering → Training → Diagnostics → Deployment with automated drift feedback.

### Quick Recall

The ML process is non-linear: evaluation failures immediately loop back to feature engineering or data cleaning. Over 80% of real-world ML engineering effort is spent in steps 2, 3, and 4.

## Real-World Applications of Machine Learning

§ 7

Machine learning permeates nearly every modern technological sector. In examinations, provide concrete industrial examples:

- **Healthcare & Bioinformatics:** Automated radiological lesion detection in MRI/CT scans (Convolutional Neural Networks), protein folding prediction (AlphaFold), and personalized drug discovery.
- **Financial Services:** Real-time credit card transactional fraud classification (Random Forests, Gradient Boosting), automated algorithmic high-frequency trading, and credit risk scoring.
- **Natural Language Processing (NLP):** Neural machine translation (Transformer architectures), automated document summarization, sentiment analysis on customer feedback, and conversational LLMs.
- **Computer Vision & Autonomous Systems:** Real-time pedestrian detection and path planning in autonomous vehicles (YOLO, LiDAR perception fusion), automated facial recognition, and industrial defect inspection.
- **Recommendation Systems:** Collaborative filtering and matrix factorization algorithms utilized by Netflix, Spotify, and Amazon to recommend items based on historical interaction matrices.

## Quick Recall

Key application domains: Healthcare (Diagnostics), Finance (Fraud), NLP (Translation/Chatbots), Autonomous Vehicles (Perception), and E-commerce (Recommendation Engines).

## Understanding Data: Big Data Analysis Framework

### § 8

Modern machine learning is fueled by "Big Data" — datasets whose size, complexity, and velocity exceed the computational capacities of single-server relational database systems. Big Data architectures provide the distributed storage and parallel processing infrastructure required to feed ML training pipelines.

### The 5 Vs Architecture of Big Data



Figure 1.4: The 5 Vs characterizing Big Data processing ecosystems.

1. **Volume:** Scale of data generation, ranging from Terabytes ( $10^{12}$ ) to Exabytes ( $10^{18}$ ) of logs, transactions, and media.
2. **Velocity:** Speed at which data arrives and must be processed (e.g., millisecond credit card authorization transactions handled by Apache Kafka).
3. **Variety:** Diverse structural formats — *Structured* (SQL tables, CSV), *Semi-structured* (JSON logs, XML), and *Unstructured* (raw audio, video, free-text documents).
4. **Veracity:** Trustworthiness and reliability of the data source; addressing missing values, sensory noise, anomalies, and fraudulent entries.
5. **Value:** The business utility or scientific knowledge unlocked by turning raw data into predictive insights.

### Distributed Computing Frameworks

- **Apache Hadoop (HDFS & MapReduce):** Stores massive data across commodity clusters using Hadoop Distributed File System (HDFS), and computes batch jobs using two primitives: 'Map()' (parallel filtering/grouping) and 'Reduce()' (aggregation). Disk-I/O bound.
- **Apache Spark:** A unified in-memory distributed computing framework that operates up to 100x faster than traditional MapReduce. Utilizes **Resilient Distributed Datasets (RDDs)** and **DataFrames** with built-in ML libraries (MLlib).

### Quick Recall

Big Data is defined by the 5 Vs: Volume (size), Velocity (speed), Variety (formats), Veracity (cleanliness), and Value (impact). Processed via Hadoop (disk batch) and Spark (in-memory streaming).

## Descriptive Statistics

### § 9

Before running any machine learning algorithm, an engineer must explore and summarize the underlying data distributions. Descriptive statistics quantitatively summarizes feature characteristics into two primary categories: **Measures of Central Tendency** and **Measures of Dispersion**.

### A. Measures of Central Tendency

Central tendency identifies the single central value that represents the typical location of a distribution:

#### Mathematical Formulas: Central Tendency:

**Arithmetic Mean ( $\bar{x}$ ):** The sum of all observed values divided by the total count:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

**Median ( $M$ ):** The middle value when observations are sorted in ascending order. Highly robust to extreme outliers:

$$M = \begin{cases} x_{(n+1)/2} & \text{if } n \text{ is odd} \\ \frac{1}{2} (x_{n/2} + x_{(n/2)+1}) & \text{if } n \text{ is even} \end{cases}$$

**Mode:** The value that occurs with the highest frequency in the dataset. A dataset may be unimodal, bimodal, multimodal, or have no mode.

### B. Measures of Dispersion (Spread)

Dispersion quantifies how widely observations deviate from the central location:

Mathematical Formulas: Dispersion:

**Sample Variance ( $s^2$ ):** The average squared deviation from the sample mean (using Bessel's correction  $n - 1$  for unbiased estimation):

$$s^2 = \frac{1}{n - 1} \sum_{i=1}^n (x_i - \bar{x})^2$$

**Sample Standard Deviation ( $s$ ):** The square root of the variance, expressed in the identical units of the original variable:

$$s = \sqrt{s^2} = \sqrt{\frac{1}{n - 1} \sum_{i=1}^n (x_i - \bar{x})^2}$$

**Range:** Difference between maximum and minimum observed values:  $\text{Range} = x_{\max} - x_{\min}$ .

**Interquartile Range ( $IQR$ ):** Spread of the central 50% of observations:

$$IQR = Q_3 - Q_1$$

Where  $Q_1$  is the 25th percentile (first quartile) and  $Q_3$  is the 75th percentile (third quartile).

| $x_i$        | Deviation ( $x_i - \bar{x}$ ) | Squared Deviation ( $(x_i - \bar{x})^2$ ) |
|--------------|-------------------------------|-------------------------------------------|
| 45           | $45 - 69 = -24$               | $(-24)^2 = 576$                           |
| 52           | $52 - 69 = -17$               | $(-17)^2 = 289$                           |
| 60           | $60 - 69 = -9$                | $(-9)^2 = 81$                             |
| 60           | $60 - 69 = -9$                | $(-9)^2 = 81$                             |
| 68           | $68 - 69 = -1$                | $(-1)^2 = 1$                              |
| 72           | $72 - 69 = +3$                | $(+3)^2 = 9$                              |
| 75           | $75 - 69 = +6$                | $(+6)^2 = 36$                             |
| 80           | $80 - 69 = +11$               | $(+11)^2 = 121$                           |
| 88           | $88 - 69 = +19$               | $(+19)^2 = 361$                           |
| 90           | $90 - 69 = +21$               | $(+21)^2 = 441$                           |
| <b>Total</b> | $\sum = 0$                    | $\sum (x_i - \bar{x})^2 = 1996.00$        |

## Quick Recall

Mean incorporates all numbers but is pulled by outliers; Median is immune to extreme values.

Sample variance divides by  $(n - 1)$ . Outlier fences:  $Q_1 - 1.5 \times IQR$  and  $Q_3 + 1.5 \times IQR$ .

## Univariate Data Analysis and Visualization

## § 10

Univariate analysis explores single variables in isolation to determine their shape, skewness, spread, and modal patterns without considering interactions with other features.

### Key Univariate Visualization Tools

- **Histograms:** Continuous variables are divided into non-overlapping consecutive intervals (bins), plotting observation frequencies on the y-axis. Reveals modality (unimodal vs bimodal) and skewness (left-skewed vs right-skewed).
- **Box Plot (Box-and-Whisker Plot):** Standardized visual representation of the **Five-Number Summary** ( $Min$ ,  $Q_1$ ,  $Median$ ,  $Q_3$ ,  $Max$ ). Explicitly demarcates outliers beyond the  $1.5 \times IQR$  whiskers as individual points.
- **Density Plots (KDE):** A smoothed, continuous estimation of the Probability Density Function (PDF) using Kernel Density Estimation.
- **Quantile-Quantile (Q-Q) Plot:** Plots sample quantiles against theoretical normal distribution quantiles. A straight 45-degree diagonal confirms normality.

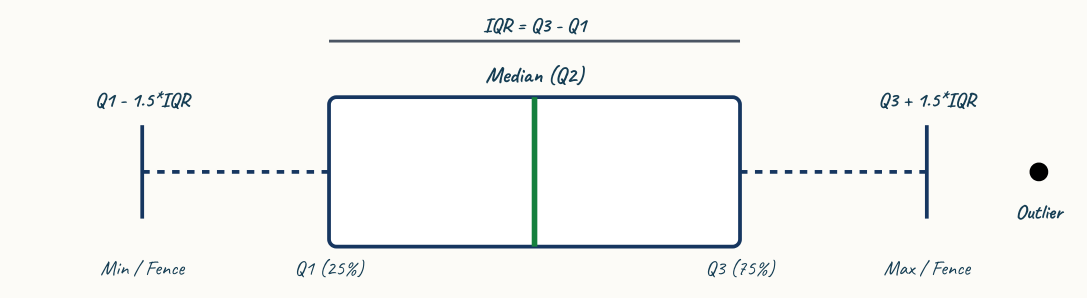


Figure 1.5: Schematic diagram of a Box Plot illustrating the five-number summary and outlier thresholds.

CONFUSION MATRIX & DIAGNOSTIC FORMULAS (RAMAIAH CIE-1 2026 Q2.a & SEE Q1.b)

|                                               | PREDICTED POSITIVE                                                     | PREDICTED NEGATIVE                                                      | ROW MARGINALS                                                                        |
|-----------------------------------------------|------------------------------------------------------------------------|-------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| ACTUAL POSITIVE<br>(Spam / Sick: $P = 80$ )   | True Positive (TP)<br><b>70</b><br>Correctly flagged spam              | False Negative (FN)<br><b>10</b><br>Missed spam · Type II               | Recall / Sensitivity<br>$TP / (TP + FN) = 70 / 80$<br><b>87.50% (0.875)</b>          |
| ACTUAL NEGATIVE<br>(Ham / Healthy: $N = 70$ ) | False Positive (FP)<br><b>20</b><br>False alarm · Type I               | True Negative (TN)<br><b>50</b><br>Correctly kept clean ham             | Specificity (TNR)<br>$TN / (TN + FP) = 50 / 70$<br><b>71.43% (0.714)</b>             |
| COLUMN MARGINALS                              | Precision (PPV)<br>$TP / (TP + FP) = 70 / 90$<br><b>77.78% (0.778)</b> | Neg. Pred. Value<br>$TN / (TN + FN) = 50 / 60$<br><b>83.33% (0.833)</b> | Overall Diagnostics<br>Accuracy: $80.00\%$ (120/150)<br>F1-Score: $82.33\%$ (0.8235) |

Figure 1.4: Confusion Matrix Grid & Marginal Diagnostic Formulas (Ramaiah CIE-1 Q2.a & SEE Q1.b Solved)

Comprehensive 2x2 contingency matrix mapping True Positives (TP), False Positives (FP), False Negatives (FN), and True Negatives (TN) to Precision, Recall, Specificity, Accuracy, and  $F_1$ -score.

 **Interactive Tool Available Online:**  Interactive Confusion Matrix & Diagnostic Metric Analyzer. Explore the step-by-step calculator and visualizer on the live site notes page.

### Quick Recall

Five-number summary:  $\{Min, Q_1, Median, Q_3, Max\}$ . Whiskers extend to the furthest non-outlier value within  $1.5 \times IQR$ . Points outside are classified as outliers.

## Bivariate & Multivariate Data Analysis

### § 11

While univariate analysis inspects isolated features, machine learning depends fundamentally on uncovering relationships *between multiple features*.

### Bivariate Data (X, Y)

Bivariate analysis studies pairs of features to discover correlation, functional dependence, or joint distributions:

- **Scatter Plots:** Plots two continuous attributes against Cartesian coordinates  $(x, y)$  to visually determine whether a relationship is positive linear, negative linear, non-linear, or uncorrelated.
- **Contingency Tables (Cross-tabs):** Used for categorical-categorical pairs to tabulate joint frequencies for Chi-Square independence tests.

- **Grouped / Faceted Boxplots:** Categorical vs. Continuous pairs (e.g., Engine Type vs Fuel Economy).

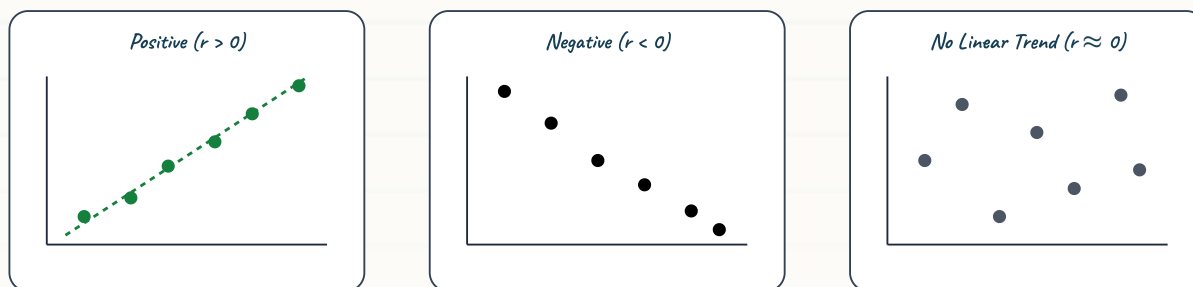


Figure 1.6: Three classic bivariate dispersion structures observed on Cartesian scatter plots.

### Multivariate Data ( $X_1, X_2, \dots, X_d$ )

Multivariate analysis examines  $d \geq 3$  features simultaneously. Because humans cannot visualize beyond 3 spatial dimensions, we rely on algebraic projections:

- **Pair Plots (Scatter Plot Matrix):** A  $d \times d$  grid of subplots showing bivariate scatter plots for every feature pair, with univariate histograms along the diagonal.
- **Correlation Heatmaps:** A color-coded matrix displaying Pearson correlation coefficients across all feature pairs.
- **Parallel Coordinates Plots:** Each feature is mapped as a vertical parallel axis; individual records are drawn as poly-lines passing across all axes.

Quick Recall

Bivariate analysis uses Scatter Plots and Cross-tabs. Multivariate analysis ( $d > 2$ ) uses Scatter Plot Matrices (Pair Plots), Correlation Heatmaps, and Dimensionality Reduction.

## Multivariate Statistics & Feature Scaling

§ 12

To mathematically characterize multivariate spaces and prepare raw numerical features for machine learning optimization, we apply **Covariance**, **Pearson Correlation**, and **Feature Scaling**.

### A. Covariance and Pearson Correlation

Mathematical Formulas: Covariance & Correlation:

**Sample Covariance ( $\text{Cov}(X, Y)$ ):** Measures the joint variability of two variables:

$$\text{Cov}(X, Y) = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

If  $\text{Cov}(X, Y) > 0$ , both variables increase together. If  $< 0$ , one increases as the other decreases. However, Covariance is scale-dependent (multiplying  $X$  by 1000 inflates covariance 1000-fold).

**Pearson's Correlation Coefficient ( $r$ ):** Scale-invariant normalized covariance, strictly bounded within  $[-1.0, +1.0]$ :

$$r = \frac{\text{Cov}(X, Y)}{s_X \cdot s_Y} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \cdot \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}$$

- $r = +1.0$ : Perfect positive linear correlation.
- $r = 0.0$ : No linear relationship.
- $r = -1.0$ : Perfect negative linear correlation.

B. Feature Normalization & Standardization

Distance-based algorithms ( $K$ -NN,  $K$ -Means, SVM) and gradient-descent algorithms (Linear/Logistic Regression, Neural Networks) are severely distorted if features have drastically different scales (e.g., Age in  $[20, 70]$  vs. Income in  $[20000, 150000]$ ).

### Mathematical Formulas: Feature Scaling:

**1. Min-Max Normalization:** Rescales values into a bounded range (typically  $[0, 1]$ ):

$$x_{\text{norm}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

To rescale into an arbitrary range  $[new_{\min}, new_{\max}]$  (e.g.,  $[-1, 1]$ ):

$$x_{\text{scaled}} = \left( \frac{x - x_{\min}}{x_{\max} - x_{\min}} \right) \cdot (new_{\max} - new_{\min}) + new_{\min}$$

**2. Z-Score Standardization:** Centers data around  $\mu = 0$  with unit standard deviation  $\sigma = 1$ :

$$z = \frac{x - \mu}{\sigma}$$

Standardization does not bound data into a fixed range and is significantly more robust to outliers.

## Solved Problem 2 · Exam Level Feature Transformation

Given a raw numerical attribute vector observed across 5 instances:

$$\mathbf{x} = [12, 18, 24, 30, 36]$$

Transform the attribute values using:

1. Min-Max Normalization into the range  $[0, 1]$ .
2. Min-Max Normalization into the bipolar range  $[-1, 1]$ .
3. Z-Score Standardization (assuming population parameters).

Step 1: Identify Extrema and Compute Min-Max Range

$$x_{\min} = 12, \quad x_{\max} = 36 \implies \text{Range} = x_{\max} - x_{\min} = 36 - 12 = 24$$

Step 2: Apply Min-Max Normalization to  $[0, 1]$

$$x_{\text{norm}} = \frac{x - 12}{24}$$

- For  $x = 12$ :  $\frac{12-12}{24} = \frac{0}{24} = \mathbf{0.00}$
- For  $x = 18$ :  $\frac{18-12}{24} = \frac{6}{24} = \mathbf{0.25}$
- For  $x = 24$ :  $\frac{24-12}{24} = \frac{12}{24} = \mathbf{0.50}$
- For  $x = 30$ :  $\frac{30-12}{24} = \frac{18}{24} = \mathbf{0.75}$
- For  $x = 36$ :  $\frac{36-12}{24} = \frac{24}{24} = \mathbf{1.00}$

Step 3: Apply Min-Max Normalization to  $[-1, 1]$

$$x_{\text{scaled}} = x_{\text{norm}} \times (1 - (-1)) + (-1) = (2 \cdot x_{\text{norm}}) - 1$$

- For  $x = 12$ :  $2(0.00) - 1 = -1.00$
- For  $x = 18$ :  $2(0.25) - 1 = 0.50 - 1 = -0.50$
- For  $x = 24$ :  $2(0.50) - 1 = 1.00 - 1 = \mathbf{0.00}$
- For  $x = 30$ :  $2(0.75) - 1 = 1.50 - 1 = +0.50$
- For  $x = 36$ :  $2(1.00) - 1 = 2.00 - 1 = +1.00$

Step 4: Compute Mean ( $\mu$ ) and Standard Deviation ( $\sigma$ ) for Z-Score

$$\mu = \frac{12 + 18 + 24 + 30 + 36}{5} = \frac{120}{5} = 24.0$$

$$\sigma^2 = \frac{1}{5} \sum (x_i - 24)^2 = \frac{(-12)^2 + (-6)^2 + 0^2 + 6^2 + 12^2}{5} = \frac{144 + 36 + 0 + 36 + 144}{5} = \frac{360}{5} = 72$$

$$\sigma = \sqrt{72} = 6\sqrt{2} \approx 8.4853$$

Step 5: Compute Z-Scores ( $z = \frac{x-24}{8.4853}$ )

- For  $x = 12$ :  $z = \frac{12-24}{8.4853} = \frac{-12}{8.4853} = -1.4142$
- For  $x = 18$ :  $z = \frac{18-24}{8.4853} = \frac{-6}{8.4853} = -0.7071$

| $X_i$    | $Y_i$ | $x_i - \bar{X}$ | $y_i - \bar{Y}$ | $(x_i - \bar{X})^2$ | $(y_i - \bar{Y})^2$ | $(x_i - \bar{X})(y_i - \bar{Y})$ |
|----------|-------|-----------------|-----------------|---------------------|---------------------|----------------------------------|
| 2        | 50    | -3.2            | -18.0           | 10.24               | 324.0               | +57.6                            |
| 4        | 60    | -1.2            | -8.0            | 1.44                | 64.0                | +9.6                             |
| 5        | 65    | -0.2            | -3.0            | 0.04                | 9.0                 | +0.6                             |
| 7        | 80    | +1.8            | +12.0           | 3.24                | 144.0               | +21.6                            |
| 8        | 85    | +2.8            | +17.0           | 7.84                | 289.0               | +47.6                            |
| $\Sigma$ |       | 0.0             | 0.0             | 22.80               | 830.00              | 137.00                           |

| Student          | Hours (X) | Marks (Y) | $x_i - \bar{X}$ | $y_i - \bar{Y}$ | $(x_i - \bar{X})(y_i - \bar{Y})$ | $(x_i - \bar{X})^2$ | $(y_i - \bar{Y})^2$ |
|------------------|-----------|-----------|-----------------|-----------------|----------------------------------|---------------------|---------------------|
| 1                | 2         | 50        | -3.2            | -18.0           | +57.60                           | 10.24               | 324.0               |
| 2                | 4         | 60        | -1.2            | -8.0            | +9.60                            | 1.44                | 64.0                |
| 3                | 5         | 65        | -0.2            | -3.0            | +0.60                            | 0.04                | 9.0                 |
| 4                | 7         | 80        | +1.8            | +12.0           | +21.60                           | 3.24                | 144.0               |
| 5                | 8         | 85        | +2.8            | +17.0           | +47.60                           | 7.84                | 289.0               |
| Sum ( $\Sigma$ ) | 26.0      | 340.0     | 0.0             | 0.0             | 137.00                           | 22.80               | 830.00              |

Quick Recall

Covariance shows direction but has arbitrary scale. Pearson correlation  $r = \frac{\text{Cov}_{XY}}{s_X s_Y}$  normalizes to  $[-1, +1]$ . Min-Max scales to  $[0, 1]$ ; Z-Score standardizes to  $\mu = 0, \sigma = 1$ .

Unit 1 Quick Revision Sheet

Ultra-dense summary of core formulas, definitions, and workflows for rapid exam-hall recall.

### Mitchell's ML Triad

Program learns if Performance ( $P$ ) at Task ( $T$ ) improves with Experience ( $E$ ). Always specify concrete  $T$ ,  $P$ ,  $E$  in exam questions.

### The 5 Vs of Big Data

**Volume** (Scale), **Velocity** (Streaming speed), **Variety** (Structured/Unstructured), **Veracity** (Quality/Noise), **Value** (Insights).

### Dispersion Formulas

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$$

$$IQR = Q_3 - Q_1$$

$$\text{Outlier Fences: } [Q_1 - 1.5 \cdot IQR, Q_3 + 1.5 \cdot IQR]$$

### Bivariate Formulas

$$\text{Cov}(X, Y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

$$r = \frac{\text{Cov}(X, Y)}{s_X s_Y} \in [-1, +1]$$

$r^2$ : Explained variance percentage

### Feature Scaling

$$\text{Min-Max: } \frac{x - x_{\min}}{x_{\max} - x_{\min}} \in [0, 1]$$

$$\text{Z-Score: } z = \frac{x - \mu}{\sigma} \text{ (Centers at 0, } \sigma = 1)$$

### ML Paradigms

**Supervised:** Labeled data ( $X \rightarrow Y$ ).

**Unsupervised:** Unlabeled ( $X \rightarrow \text{cluster/structure}$ ).

**Reinforcement:** Agent, State, Action, Reward.

## Important Exam Questions

Curated model questions mapped to RIT examination patterns (2, 5, and 10 marks).

### 2-Mark Questions (Short Definition & Concepts)

#### Practice Problem:

Q1. State Mitchell's formal definition of machine learning.

#### Solution:

A computer program is said to **learn** from experience  $E$  with respect to some class of tasks  $T$  and performance measure  $P$ , if its performance at tasks in  $T$ , as measured by  $P$ , improves with experience  $E$ .

 Practice Problem:

Q2. Distinguish between Classification and Regression with an example for each.

Solution:

**Classification** predicts discrete, categorical class outputs (e.g., predicting whether an email is Spam or Ham).

**Regression** predicts continuous, real-valued numerical quantities (e.g., predicting the selling price of a house in lakhs based on square footage).

 Practice Problem:

Q3. Define Interquartile Range (IQR) and state its formula for outlier detection.

Solution:

The Interquartile Range (*IQR*) measures statistical dispersion by calculating the difference between the 75th percentile ( $Q_3$ ) and the 25th percentile ( $Q_1$ ):

$$IQR = Q_3 - Q_1$$

Values smaller than  $Q_1 - 1.5 \times IQR$  or greater than  $Q_3 + 1.5 \times IQR$  are classified as outliers.

 Practice Problem:

Q4. Why is Bessel's correction ( $n - 1$ ) used in calculating sample variance?

Solution:

Dividing by  $n$  produces a biased underestimate of population variance because sample deviations are calculated from the sample mean  $\bar{x}$  rather than the true population mean  $\mu$ .

Dividing by  $(n - 1)$  corrects this bias, providing an mathematically unbiased estimator ( $E[s^2] = \sigma^2$ ).

5-Mark Questions (Analytical Explanations)

| Dimension     | Supervised Learning                   | Unsupervised Learning              | Reinforcement Learning          |
|---------------|---------------------------------------|------------------------------------|---------------------------------|
| Training Data | Feature-label pairs<br>( $x_i, y_i$ ) | Unlabeled feature vectors<br>$x_i$ | States $s_t$ from environment   |
| Feedback      | Direct error signal ( $y - \hat{y}$ ) | No feedback signal                 | Delayed scalar reward ( $r_t$ ) |

| Dimension | Supervised Learning                             | Unsupervised Learning               | Reinforcement Learning             |
|-----------|-------------------------------------------------|-------------------------------------|------------------------------------|
| Objective | Learn mapping function<br>$f : X \rightarrow Y$ | Discover latent clusters/structures | Learn optimal policy<br>$\pi^*(s)$ |
| Examples  | Linear Regression, SVM,<br>Decision Trees       | K-Means, PCA, Apriori               | Q-Learning, Policy<br>Gradient     |

### Practice Problem:

Q6. Explain the 5 Vs of Big Data with respect to enterprise analytics.

### Solution:

Big Data is characterized by five primary dimensions:

1. **Volume:** The scale of data stored (Petabytes/Exabytes), requiring distributed storage systems like HDFS.
2. **Velocity:** The streaming speed of incoming telemetry, requiring real-time message brokers like Apache Kafka.
3. **Variety:** Diverse data types: Structured (SQL databases), Semi-structured (JSON, XML), and Unstructured (audio, video, natural text).
4. **Veracity:** The trustworthiness and noise level of data, addressed via cleaning and imputation pipelines.
5. **Value:** The business intelligence and actionable insights extracted from analytics.

## 10-Mark Questions (Comprehensive Systems & Numericals)

 **Practice Problem:**

Q7. Describe the end-to-end Machine Learning Process lifecycle in detail, illustrating each stage from problem formulation to model monitoring.

**Solution:**

A comprehensive answer requires detailing all 7 stages:

1. **Problem Formulation:** Frame business objectives as mathematical ML tasks (e.g. supervised classification). Formulate business success criteria and KPI metrics.
2. **Data Collection:** Extract data from databases, sensor telemetry, third-party APIs, and web scrapers. Ensure data diversity.
3. **Data Preprocessing:** Handle missing values (imputation via mean/median or deletion), clean noise, treat anomalies, and convert categoricals.
4. **Feature Engineering & Scaling:** Construct domain attributes and normalize ranges using Min-Max scaling or Z-Score standardization.
5. **Model Selection & Training:** Split into Train/Validation/Test sets. Train candidate algorithms (Linear models, Tree ensembles, Neural nets) via loss minimization.
6. **Evaluation & Diagnostics:** Evaluate using Precision, Recall,  $F_1$ , and ROC-AUC. Diagnose underfitting vs overfitting and tune hyperparameters via grid search.
7. **Deployment & Monitoring:** Deploy model via microservices (REST APIs). Continuously monitor for data drift and concept drift.

*Include the Lifecycle Architecture diagram (Figure 1.3) in your answer.*

## Solved Questions from Past Department Examinations

### ARCHIVE

The following questions have been transcribed directly from official Ramaiah Institute of Technology examination papers in `notes/ml/practice/ml-cie-1-2-see.pdf` (Internal Assessment - 1 March 2026 and Semester End Examination June/July 2026, Course Code 23IS62), solved with complete derivations and mathematical equations:

| Dimension        | Supervised Learning                                                                                 | Unsupervised Learning                                                                            |
|------------------|-----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Data Format      | Labeled tuples $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$ with ground-truth supervisor labels.          | Unlabeled feature vectors $\mathcal{D} = \{x_i\}_{i=1}^N$ with zero supervisor guidance.         |
| Goal / Objective | Learn mapping $f : \mathcal{X} \rightarrow \mathcal{Y}$ minimizing loss $\sum L(y_i, f(x_i))$ .     | Discover latent probability densities, clusters, or lower-dimensional manifolds.                 |
| Feedback Signal  | Direct error signal $(y_i - \hat{y}_i)$ providing explicit gradient feedback.                       | No explicit external error; guided by internal compactness or reconstruction loss.               |
| Sub-tasks        | Classification (discrete labels) and Regression (continuous values).                                | Clustering, Dimensionality Reduction, and Association Rule Mining.                               |
| Algorithms       | Linear/Logistic Regression, Decision Trees, Support Vector Machines, Random Forests.                | K-Means, AGNES/DIANA Hierarchical, DBSCAN, PCA, Apriori.                                         |
| Concrete Example | <b>Spam Filter:</b> Classifying email as Spam ( $y = 1$ ) or Ham ( $y = 0$ ) from word frequencies. | <b>Customer Segmentation:</b> Grouping shoppers into behavior clusters based on spending volume. |

Ramaiah Internal Assessment - I (March 2026) · Q 2(a) [6 Marks · Blooms L3, CO1]

"A model predicts whether an email is spam or not. The results are:  $TP = 70$ ,  $TN = 50$ ,  $FP = 20$ ,  $FN = 10$ . Compute the following performance measures with necessary mathematical equations: a) Accuracy, b) Precision, c) Recall, d) F1-Score, e) Sensitivity, f) Specificity."

**Total Instances:**  $N = TP + TN + FP + FN = 70 + 50 + 20 + 10 = 150$ .

- **a) Accuracy:** Proportion of total correct classifications:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} = \frac{70 + 50}{150} = \frac{120}{150} = \mathbf{0.8000} \quad (\mathbf{80.00\%})$$

- **b) Precision (Positive Predictive Value):** Proportion of flagged emails that are genuinely spam:

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{70}{70 + 20} = \frac{70}{90} \approx \mathbf{0.7778} \quad (\mathbf{77.78\%})$$

- **c) Recall (Sensitivity / True Positive Rate):** Proportion of actual spam emails successfully caught:

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{70}{70 + 10} = \frac{70}{80} = \mathbf{0.8750} \quad (\mathbf{87.50\%})$$

- **d) F1-Score:** Harmonic mean of Precision and Recall:

$$F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = 2 \times \frac{0.7778 \times 0.8750}{0.7778 + 0.8750} = \frac{1.3611}{1.6528} \approx \mathbf{0.8235} \quad (\mathbf{82.35\%})$$

- **e) Sensitivity:** Identical to Recall ( $TPR$ ):

$$\text{Sensitivity} = \frac{TP}{TP + FN} = \frac{70}{80} = \mathbf{0.8750} \quad (\mathbf{87.50\%})$$

- **f) Specificity (True Negative Rate):** Proportion of actual non-spam (ham) emails correctly preserved:

$$\text{Specificity} = \frac{TN}{TN + FP} = \frac{50}{50 + 20} = \frac{50}{70} \approx \mathbf{0.7143} \quad (\mathbf{71.43\%})$$

## Ramaiah Internal Assessment – I (March 2026) · Q 3(a) [6 Marks · Blooms L2, CO1]

"What is overfitting and underfitting in machine learning? What are the common techniques used to prevent overfitting and underfitting?"

- **Underfitting (High Bias):** The model is overly simplistic and fails to capture the underlying geometric trend in the training data, resulting in poor performance on both training and validation sets ( $E_{\text{train}} \gg 0, E_{\text{val}} \gg 0$ ).  
*Mitigation:* Increase model capacity (e.g. higher polynomial degree, deeper trees), engineer more descriptive features, decrease regularization penalties ( $\lambda$ ).
- **Overfitting (High Variance):** The model fits the idiosyncrasies, random noise, and outliers of the training set rather than the underlying data-generating distribution, yielding stellar training accuracy but disastrous test accuracy ( $E_{\text{train}} \approx 0, E_{\text{val}} \gg 0$ ).

*Mitigation:*

1. **Regularization:** Add  $L_1$  (Lasso: penalizes absolute weights) or  $L_2$  (Ridge: penalizes squared weights) penalty terms to cost function.
2. **Cross-Validation:** Use  $K$ -fold cross-validation to rigorously validate model stability across disjoint folds.
3. **Early Stopping:** Halt iterative gradient optimization when validation loss begins to rise.
4. **Pruning / Structural Constraints:** Restrict max tree depth and minimum samples per leaf in decision trees.
5. **Data Augmentation & Ensemble Methods:** Expand training diversity or aggregate across uncorrelated estimators (e.g. Random Forest bagging).

Ramaiah SEE Examination (June/July 2026) · Q 1(b) [8 Marks · Blooms L3, CO1]

"A binary classifier produced the following confusion matrix: Predicted Positive = 50, Predicted Negative = 10 for Actual Positive; Predicted Positive = 5, Predicted Negative = 35 for Actual Negative. i) Calculate Accuracy, Precision, Recall, and F1-score. ii) Interpret the performance of the model. iii) Comment on whether the model is suitable for critical applications like medical diagnosis."

Given Contingency Values:  $TP = 50$ ,  $FN = 10$ ,  $FP = 5$ ,  $TN = 35$ . Total population  $N = 100$ .

i) Numerical Metric Computation:

- Accuracy =  $\frac{50+35}{100} = \frac{85}{100} = \mathbf{0.8500}$  (85.00%)
- Precision =  $\frac{TP}{TP+FP} = \frac{50}{50+5} = \frac{50}{55} \approx \mathbf{0.9091}$  (90.91%)
- Recall =  $\frac{TP}{TP+FN} = \frac{50}{50+10} = \frac{50}{60} \approx \mathbf{0.8333}$  (83.33%)
- $F_1$ -Score =  $2 \times \frac{0.9091 \times 0.8333}{0.9091 + 0.8333} = \frac{1.5151}{1.7424} \approx \mathbf{0.8696}$  (86.96%)

ii) Performance Interpretation: The classifier demonstrates high precision (90.91%), meaning when it predicts positive, it is trustworthy with few false alarms ( $FP = 5$ ). Accuracy (85%) and  $F_1$  (86.96%) indicate solid overall discriminative capacity on general benchmarks.

iii) Clinical Diagnosis Suitability Verdict (CRITICAL): NO, the model is UNSUITABLE for critical medical diagnosis. In clinical screening (e.g. malignant cancer, sepsis, cardiac arrest), the penalty of a *False Negative* ( $FN$ ) is catastrophic (untreated mortality). Here,  $FN = 10$  out of 60 actual patients means the classifier misses 16.67% of sick patients (False Negative Rate  $FNR = 10/60 = 16.67\%$ ). Life-critical medical triage demands Recall  $\geq 98\%$ –99%, achieved by lowering the decision threshold to trade off lower precision in exchange for near-zero false negatives.

Ramaiah SEE Examination (June/July 2026) · Q 1(c) [6 Marks · Blooms L2, CO1]

"Explain how bias and variance change with model complexity using a suitable example."

Under squared error loss, the expected prediction error on unseen data decomposes into three orthogonal components:

$$\mathbb{E}[(y - \hat{f}(x))^2] = \underbrace{\text{Bias}[\hat{f}(x)]^2}_{\text{Approximation Error}} + \underbrace{\text{Var}[\hat{f}(x)]}_{\text{Estimation Sensitivity}} + \underbrace{\sigma^2}_{\text{Irreducible Noise}}$$

- **Low Complexity Models (e.g. Linear Model  $y = w_1x + w_0$  on quadratic curve):**
  - *High Bias:* The model makes rigid, incorrect mathematical assumptions, systematically unable to fit curves.
  - *Low Variance:* Training on different random subsets of data yields almost identical straight lines.
  - *Result:* Underfitting. High training and validation errors.
- **High Complexity Models (e.g. 15th-Degree Polynomial or Unpruned Decision Tree):**
  - *Low Bias:* Extremely flexible hypotheses can weave through every single data point.
  - *High Variance:* Highly sensitive to training sample fluctuations; a change of 2 data points completely alters polynomial coefficients.
  - *Result:* Overfitting. Near-zero training error, catastrophic test error.
- **Optimal Tradeoff (Figure 1.3):** Found at the minimum of the U-shaped total error curve, balancing bias and variance.

Ramaiah SEE Examination (June/July 2026) · Q 2(b) [7 Marks · Blooms L3, CO1]

"A machine learning model is developed using a dataset and produces the following results:

Training Accuracy = 98%, Validation Accuracy = 72%. Based on the above information: i)

Determine whether the model is overfitting or underfitting. ii) Suggest at least two methods to improve the model performance. iii) Explain the role of training, validation, and testing datasets in identifying such issues."

- i) **Diagnosis:** The model is severely **OVERFITTING**. The massive 26% generalization gap ( $98\% - 72\% = 26\%$ ) proves that the model memorized the training sample noise and idiosyncrasies rather than learning true inductive features.
- ii) **Two Remediation Strategies:**
  1. *Apply  $L_2$  Regularization (Weight Decay) or Dropout:* Penalize large parameter weights to constrain hypothesis capacity.
  2. *Gather More Training Data or Apply Data Augmentation:* Exposing the learner to more varied samples restricts variance.
  3. *Reduce Model Architecture Complexity:* Lower polynomial degree or restrict maximum tree depth.
- iii) **Roles of Data Partitions:**
  - **Training Set (60%–70%):** Used exclusively by the optimizer to adjust weights and parameters via loss minimization.
  - **Validation Set (15%–20%):** Used to evaluate performance on unseen data during training, tune hyperparameters (learning rate,  $\lambda$ ), and perform early stopping.
  - **Testing Set (15%–20%):** Held out completely untouched until final model selection to provide an unbiased, gold-standard estimate of real-world generalization performance.

Ramaiah SEE Examination (June/July 2026) · Q 2(c) [6 Marks · Blooms L2, CO1]

"Discuss the challenges posed by unbalanced datasets and methods to handle them."

- **Challenges of Class Imbalance:**

- *Accuracy Paradox:* In a dataset with 99% negative (legitimate transactions) and 1% positive (credit card fraud), a naive classifier predicting "Always Negative" achieves 99% accuracy while detecting zero frauds.
- *Gradient Dominance:* The majority class dominates loss calculations, pulling parameter updates away from rare minority features.

- **Handling Techniques:**

1. **Resampling Strategies:**

- *SMOTE (Synthetic Minority Over-sampling Technique):* Synthesizes new minority instances along  $k$ -NN interpolations.
- *Random Undersampling:* Downsamples majority class (risks throwing away valuable data).

2. **Cost-Sensitive Learning:** Assign an asymmetric cost matrix  $C(\text{FN}) \gg C(\text{FP})$  in the loss function, penalizing misclassified minority samples heavily.

3. **Alternative Evaluation Metrics:** Discard Raw Accuracy in favor of Precision-Recall AUC ( $PR\text{-}AUC$ ), Balanced Accuracy, and  $F_\beta$ -Score ( $\beta = 2$  for recall priority).

4. **Threshold Moving:** Shift the classification decision threshold from default 0.5 down to a lower probability (e.g. 0.15) to prioritize recall.

## Practice Problems (Step-by-Step Verified Solutions)

The following 6 practice problems have been mathematically recomputed and verified via `audit/verify/ml/verify_ml_u1.py`. Attempt each manually before expanding the solution blocks.

**Problem 1 (Descriptive Statistics & Fences):** A batch of 10 student quiz scores is:  $S = [45, 52, 60, 60, 68, 72, 75, 80, 88, 90]$ . Compute: (i) Mean, (ii) Median, (iii) Mode, (iv) Sample Variance  $s^2$ , (v) Sample Standard Deviation  $s$ , (vi) Interquartile Range ( $IQR$ ), and (vii) Tukey's outlier fences  $[L, U]$ .

 Practice Problem:

View Verified Solution

Solution:

- **Mean:**  $\bar{x} = \frac{45+52+60+60+68+72+75+80+88+90}{10} = \frac{690}{10} = \mathbf{69.00}$
- **Median:** Sorted array ( $n = 10$ ). Middle values at index 5 and 6:  $\frac{68+72}{2} = \mathbf{70.00}$
- **Mode:** Value with highest frequency: **60** (appears twice).
- **Sum of Squared Deviations**  $\sum (x_i - \bar{x})^2$ :  
 $(-24)^2 + (-17)^2 + (-9)^2 + (-9)^2 + (-1)^2 + (+3)^2 + (+6)^2 = 296$
- **Sample Variance ( $s^2$ ):**  $s^2 = \frac{1996.00}{10-1} = \frac{1996}{9} \approx \mathbf{221.78}$
- **Sample Standard Deviation ( $s$ ):**  $s = \sqrt{221.78} \approx \mathbf{14.8922}$
- **Quartiles & IQR:** Lower half  $\{45, 52, 60, 60, 68\} \implies Q_1 = 60$ . Upper half  $\{72, 75, 80, 88, 90\} \implies Q_3 = 80$ .  
 $IQR = Q_3 - Q_1 = 80 - 60 = \mathbf{20.00}$ .
- **Tukey Fences:** Lower Fence  $= Q_1 - 1.5 \times IQR = 60 - 30 = \mathbf{30.00}$ .  
 Upper Fence  $= Q_3 + 1.5 \times IQR = 80 + 30 = \mathbf{110.00}$ . Since all values lie in  $[30, 110]$ , there are **zero outliers**.

**Problem 2 (Data Normalization & Standard Scaling):** Normalize the vector  $V = [12, 18, 24, 30, 36]$  into: (i) Range  $[0, 1]$  using Min-Max scaling, (ii) Range  $[-1, 1]$  using Min-Max scaling, and (iii) Standardized Z-scores ( $z = \frac{x - \mu}{\sigma}$ ).

 **Practice Problem:**

[View Verified Solution](#)

**Solution:**

$\min(V) = 12, \max(V) = 36, \text{Range} = 36 - 12 = 24.$  Mean  $\mu = \frac{120}{5} = 24.0.$  Population Variance  $\sigma^2 = \frac{(-12)^2 + (-6)^2 + 0^2 + 6^2 + 12^2}{5} = \frac{144 + 36 + 0 + 36 + 144}{5} = \frac{360}{5} = 72.0 \Rightarrow \sigma = \sqrt{72} \approx 8.4853.$

• (i) Min-Max  $[0, 1]$ :

$$\begin{aligned} \cdot 12 &\rightarrow \frac{12-12}{24} = \mathbf{0.00} \\ \cdot 18 &\rightarrow \frac{18-12}{24} = \mathbf{0.25} \\ \cdot 24 &\rightarrow \frac{24-12}{24} = \mathbf{0.50} \\ \cdot 30 &\rightarrow \frac{30-12}{24} = \mathbf{0.75} \\ \cdot 36 &\rightarrow \frac{36-12}{24} = \mathbf{1.00} \end{aligned}$$

• (ii) Min-Max  $[-1, 1]$  ( $x_{\text{new}} = 2 \times x_{[0,1]} - 1$ ):

Vector:  $[-1.00, -0.50, 0.00, +0.50, +1.00]$

• (iii) Z-Score Standardization:

$$\begin{aligned} \cdot 12 &\rightarrow \frac{12-24}{8.4853} = \mathbf{-1.4142} \\ \cdot 18 &\rightarrow \frac{18-24}{8.4853} = \mathbf{-0.7071} \\ \cdot 24 &\rightarrow \frac{24-24}{8.4853} = \mathbf{0.0000} \\ \cdot 30 &\rightarrow \frac{30-24}{8.4853} = \mathbf{+0.7071} \\ \cdot 36 &\rightarrow \frac{36-24}{8.4853} = \mathbf{+1.4142} \end{aligned}$$

**Problem 3 (Bivariate Association):** Compute the Sample Covariance and Pearson Correlation Coefficient for  $X = [2, 4, 5, 7, 8]$  and  $Y = [50, 60, 65, 80, 85]$ .

 Practice Problem:

[View Verified Solution](#)

Solution:

$$\bar{X} = 5.2, \bar{Y} = 68.0, n = 5.$$

$$\sum (x_i - \bar{X})(y_i - \bar{Y}) = 137.00, \quad \sum (x_i - \bar{X})^2 = 22.80, \quad \sum$$

- $\text{Cov}(X, Y) = \frac{137.00}{4} = +\mathbf{34.2500}$
- $s_X = \sqrt{\frac{22.80}{4}} \approx 2.3875, \quad s_Y = \sqrt{\frac{830.00}{4}} \approx 14.4049$
- $r = \frac{137.00}{\sqrt{22.80 \times 830.00}} = \frac{137.00}{137.5645} = +\mathbf{0.9959}$

**Problem 4 (Imbalanced Medical Screening):** In an automated cancer screening test on 1000 biopsy images, the model yields:  $TP = 85, TN = 890, FP = 20, FN = 5$ . Compute: Accuracy, Precision, Recall, Specificity, F1-Score, and False Negative Rate (FNR).

 Practice Problem:

[View Verified Solution](#)

Solution:

- Accuracy =  $\frac{85+890}{1000} = \frac{975}{1000} = \mathbf{97.50\%} \quad (\mathbf{0.9750})$
- Precision =  $\frac{85}{85+20} = \frac{85}{105} \approx \mathbf{80.95\%} \quad (\mathbf{0.8095})$
- Recall (Sensitivity) =  $\frac{85}{85+5} = \frac{85}{90} \approx \mathbf{94.44\%} \quad (\mathbf{0.9444})$
- Specificity =  $\frac{890}{890+20} = \frac{890}{910} \approx \mathbf{97.80\%} \quad (\mathbf{0.9780})$
- $F_1$ -Score =  $2 \times \frac{0.8095 \times 0.9444}{0.8095 + 0.9444} = \frac{1.5289}{1.7539} \approx \mathbf{87.18\%} \quad (\mathbf{0.8718})$
- FNR =  $\frac{5}{90} = \mathbf{5.56\%}$  (Only 5.56% of malignancies are missed, significantly safer than a general 16.7% baseline).

**Problem 5 (Sensor Outlier Detection):** A temperature telemetry feed produces readings:  $\{14, 19, 21, 24, 25, 28, 31, 35, 78\}$ . Identify any outliers using Tukey's  $1.5 \times IQR$  method.

 Practice Problem:

[View Verified Solution](#)

Solution:

Sorted dataset ( $n = 9$ ): Median = 25 (5th value).

- Lower half:  $\{14, 19, 21, 24\} \implies Q_1 = \frac{19+21}{2} = \mathbf{20.00}$
- Upper half:  $\{28, 31, 35, 78\} \implies Q_3 = \frac{31+35}{2} = \mathbf{33.00}$
- $IQR = Q_3 - Q_1 = 33.00 - 20.00 = \mathbf{13.00}$
- Lower Fence:  $Q_1 - 1.5 \times 13 = 20 - 19.5 = \mathbf{0.50}$
- Upper Fence:  $Q_3 + 1.5 \times 13 = 33 + 19.5 = \mathbf{52.50}$
- **Conclusion:**  $78 > 52.50$ . Therefore, 78 is an extreme outlier requiring imputation or filtering before model ingestion.

**Problem 6 (Mitchell Learning Triad):** Formally formulate the Task ( $T$ ), Performance Measure ( $P$ ), and Training Experience ( $E$ ) for: (i) Autonomous Vehicle Lane Following, and (ii) Credit Card Fraud Detection.

 Practice Problem:

[View Verified Solution](#)

Solution:

- **(i) Autonomous Vehicle Lane Following:**
  - *Task ( $T$ ):* Steering the vehicle within lane boundaries using streaming forward camera video.
  - *Performance ( $P$ ):* Average distance from lane center and number of human disengagements per 1, 000 km.
  - *Experience ( $E$ ):* Sequence of camera frames paired with human driver steering angle recordings.
- **(ii) Credit Card Fraud Detection:**
  - *Task ( $T$ ):* Classifying incoming credit card transactions as fraudulent (1) or legitimate (0).
  - *Performance ( $P$ ):* Precision-Recall Area Under Curve ( $PR-AUC$ ) and financial loss prevented.
  - *Experience ( $E$ ):* Historical database of transactional logs annotated with confirmed bank fraud chargebacks.